



MIKE SHE – Eco Lab Plant Growth

User Guide

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1 Background

The [PHISHES project](#) (PHysically-Based Integrated Soil HEalth Simulation Platform) is a 4-year European Horizon initiative developing the PHISHES Digital Platform (PDP) to predict soil health, integrate numerical modelling, and support ecosystem services.

The PDP can be accessed here: [PHISHES Digital Platform](#)

- Core goals and design: The PHISHES Digital Platform (PDP) supports the transition from soil monitoring to informed decision-making by providing access to validated test-site evidence, modelling workflows and transferable methodologies that help assess how land management, remediation and nature-based solutions may influence soil health over time.
- Timeline: Launched on September 1, 2024, and runs for 48 months through August 2028.
- Platform: Hosts data management pipelines, basin-scale model codes, and scenario tools to help decision-makers evaluate soil management options.
- Funding: Backed by the European Union under the Horizon Europe program and aligned with the EU Mission 'A Soil Deal for Europe'.

There are four European case studies:

- Czech Republic: Analysing soil management in the Želivka River Basin for drinking water quality.
- Poland: Monitoring agricultural irrigation and water withdrawals in the Koprzywianka River area.
- Sweden: Testing PFAS contamination mitigation measures at an Örnköldsvik airport site.
- Netherlands: Evaluating urban green infrastructure for resilience in the Rotterdam urban area.

The PDP offers four different types of modelling tools, also known as model trains, each producing different outputs. There are four model trains:

- MIKE SHE-Daisy: soil conditions, crop response, carbon sequestration, nutrient and contaminant leaching, catchment water balance
- MIKE SHE-ECO Lab PGM: crop yield, biomass, SOC, N cycling, water balance, ecosystem-service indicators, soil-function indicators
- MODFLOW 6-Reservoir: soil moisture, groundwater levels, infiltration, inundation, contaminant transport, urban hydrology indicators
- HYDRUS-PHREEQC-MODFLOW-MT3D: PFAS/metal fate and transport, groundwater contamination, remediation performance, pollution buffering, water balance

This user guide is intended to support the application of the MIKE SHE – Eco Lab Plant Growth model train, developed within the context of the PHISHES project – also referred to in this document as the Plant Growth Module, PGM.



2 Installation Prerequisites

The following prerequisites are required when working with the PGM in the MIKE modelling suite:

- Python: <https://www.python.org/downloads/> (Python $\geq$ 13)
- MIKE v2026: <https://support.dhigroup.com/download/MIKE-latest/>
- MIKE SHE
- MIKE ECO Lab
- uv, as Python package manager: <https://docs.astral.sh/uv/getting-started/installation/>

Users should make sure to check for hotfixes for the MIKE modelling suite before using.

Note, that python is used needed for MIKE SHE, whereas uv is used for executing the MSHE-Ecolab-PGM notebooks.



3 Assumptions

To work with the PGM, it is assumed that the users:

- Have installed all the prerequisites
- Have a running model of MIKE SHE
- Have experience with Eco Lab templates
- Download the MSHE-Ecolab-PGM from the PDP python repository:
- [phishes-pdp/tree/main/model-trains/MSHE-Ecolab-PGM](https://github.com/phishes/pdp/tree/main/model-trains/MSHE-Ecolab-PGM)

which includes 4 notebooks:

- `pgm_forcing_generator.ipynb`
 - `pgm_soil_profile_setup.ipynb`
 - `pgm_initial_condition_dfs2_map_generator.ipynb`
 - `pgm_initial_condition_updater.ipynb`
 - `result-inspection\obs_res_compare_cernici.ipynb`
 - `result-inspection\plot_mikeshe_veg_cernizi_sz.py`
- Download PGM: the 2 ECO Lab templates
 - 1 ECO Lab template for unsaturated zone
 - 1 ECO Lab template for saturated zone
 - Python scripts for python plugins in MIKE SHE
 - Download the available database of constants used in the PGM

Note: For the plant-growth-module model-train, users need to read '*README.md*' file beforehand to appropriately set up the working environment.

In this user guide, an example named 'Cernici', a small agricultural area in Czechia, is used to demonstrate.

Note that the data (e.g. land use, soil profile, etc.) here does not reflect the actual Cernici catchment model parameters.



4 Workflow Overview

MIKE SHE is coupled with the PGM for dynamic vegetation representation. The PGM is built on MIKE ECO Lab. Users should refer to the user manual for MIKE SHE (DHI, 2025) and MIKE ECO Lab (DHI, 2025) respectively to understand how these 2 models work. The following workflow is needed to set up the coupling between MIKE SHE and the PGM (see Figure 4.1).

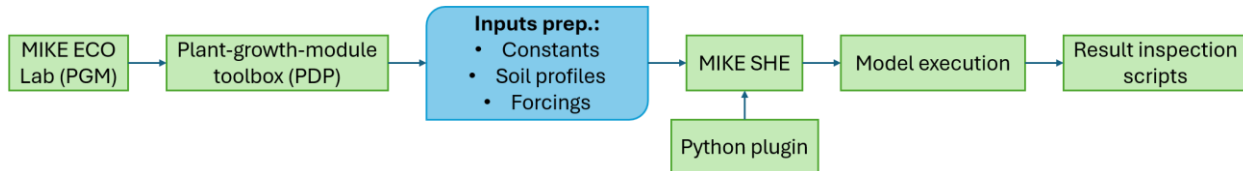


Figure 4.1 Workflow to incorporate MIKE SHE with PGM.

To incorporate the PGM in the existing MIKE SHE model, the following must be enabled in the MIKE SHE setup (see Figure 4.2):

- Enable Plugins
 - The Python plugin is specifically developed to allow for 2-way coupling between the PGM and MIKE SHE. The 2-way coupling includes: 1) calculation of state variables i.e. leaf area index, root depth, crop coefficient in the PGM using the climatic forcings from MIKE SHE; 2) transferring these parameters to MIKE SHE for calculations in the water movement module (e.g. root water uptake, transpiration, etc.)
 - The Python plugin also includes handling of properties related to soil-vegetation (e.g. soil organic carbon) during wetting and drying of the unsaturated zone from the saturated zone (i.e. varying groundwater table)
- Enable MIKE ECO Lab
 - The PGM which primarily interacts with the unsaturated zone is developed in a MIKE ECO Lab template. The ECO Lab template includes forcings and constants that need to be specified in the MIKE SHE setup. Some forcings and constants are shared with MIKE SHE (e.g. climatic forcings, soil moisture, vertical cell depth, etc.). Other forcings (e.g. seeding time for crops, fertilizers, etc.) and constants (e.g. maximum growth rate, optimal temperature for growth, etc.) need to be specified in the setup of 'Unsaturated Zone'. Initial conditions for some state variables would also need to be specified.
 - The PGM also includes a simplified process for the saturated zone, primarily for denitrification processes.

With the latest MIKE SHE development, all necessary water movement modules, including 'Rivers, Lakes, Sewers' can work with PGM.



Simulation specification

Water Movement (WM)

- ☒ Overland Flow (OL)

Finite Difference
- ☒ Rivers, Lakes, Sewers (OC)
- ☒ Unsaturated Flow (UZ)

Richards equation
- ☒ Evapotranspiration (ET)

Kristensen&Jensen
- ☒ Saturated Flow (SZ)

Finite Difference
- ☒ Enable Plugins

☒ Include Advection-Dispersion (AD) Water Quality

- ☒ Calculate WQ using the finite difference, Advection-Dispersion (AD) method
- ☐ Calculate WQ using random walk particle tracking (SZ only)
- ☒ Use current WM simulation for Water Quality

Water Quality Simulation Specification

	WM	WQ
Overland Flow (OL):	☒	☒
Rivers, Lakes and Sewers	☒	☒ Requires WQ in Overland Flow
Unsaturated Flow (UZ):	☒	☒ Richards Eq and Gravity methods only
Evapotranspiration (ET):	☒	☒ Plant uptake
Saturated Flow (SZ):	☒	☒

Water Quality Processes

- ☒ Include Water Quality Processes
 - ☐ Sorption and Decay
 - ☒ MIKE ECO Lab

Figure 4.2 Simulation specifications: Enable Plugin. Water Quality Simulation Specification: Enable MIKE ECO Lab.

- Input preparation
 - The input preparation requires using the model train 'MSHE-Ecolab-PGM' from the PDP python repository. There are 3 Python notebooks, corresponding to inputs necessary to run the PGM with MIKE SHE. These are only used for inputs in the 'Unsaturated Zone' due to the spatially varying vegetation and land-use coverage. The constants and forcings in the 'Saturated Zone' do not necessarily require input preparation outside of MIKE SHE's GUI. For usage of these Python notebooks, refer to the individual notebooks for instructions.
 - pgm_forcing_generator.ipynb: for forcings such as seeding, fertilization
 - pgm_soil_profile_setup.ipynb: for field capacity and wilting point at different soil layers
 - pgm_initial_condition_dfs2_map_generator.ipynb: for constants and initial values
 - pgm_initial_condition_updater.ipynb: for initial condition wiring from a 3D WQ result to a MIKE SHE model.
- Result processing and inspection
 - The result processing notebook (obs_res_compare_cernici .pynb) makes it easier to handle results of the PGM in comparison with observed data e.g. harvested biomass, actual evapotranspiration, etc.



5 Enable Plugin

MIKE SHE setup requires enabling of Python plugins by ticking '*Enable Plugins*' in the Simulation Specification window (see Figure 4.2). This allows for the 2-way coupling between MIKE SHE and MIKE ECO Lab. The plugin requires installation of Python ≥ 13 . Users are required to specify the path to the '*python{version}.dll*' and only need to set this up once.

The python script used for the plugin can be changed depending on the project; however, it is advised to do so with precaution.

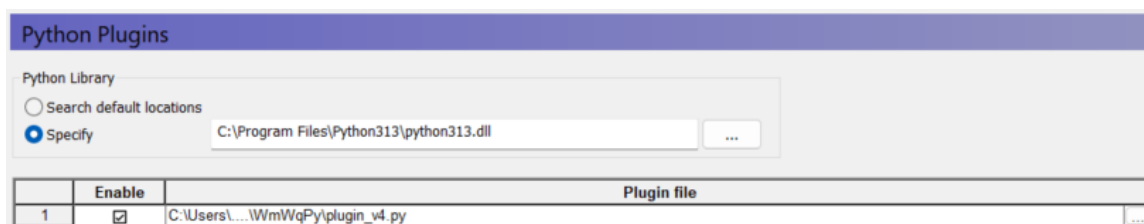


Figure 5.1 In the Python Plugins tab, specify own python location and the python script.



6 Enable MIKE ECO Lab

After selecting MIKE ECO Lab for water quality processes, choose the ECO Lab template for the PGM for the unsaturated zone. Another ECO Lab template for the saturated zone can be added in. The templates work independently, e.g. if the computational cell is saturated, the equations in the ECO Lab template for '*Saturated Zone*' are used.

MIKE ECO Lab Template Specification

Overland flow and Ground surface

☐ Enable MIKE ECO Lab for Overland flow and Ground surface

MIKE ECO Lab

...
Edit...

Integration method: EULER

Update frequency: 1

0	State	0	User Specified Constants	0	User Specified	0	Processes
0	Auxiliary	0	MIKE SHE built-in Constants	0	MIKE SHE built-in Forcings	0	Derived Output

Unsaturated zone

☒ Enable MIKE ECO Lab for Unsaturated zone

MIKE ECO Lab

...
Edit...

Integration method: EULER

Update frequency: 1

96	State	244	User Specified Constants	17	User Specified	270	Processes
129	Auxiliary	4	MIKE SHE built-in Constants	4	MIKE SHE built-in Forcings	0	Derived Output

Saturated Zone

☒ Enable MIKE ECO Lab for Saturated zone

MIKE ECO Lab

...
Edit...

Integration method: EULER

Update frequency: 1

4	State	8	User Specified Constants	0	User Specified	4	Processes
5	Auxiliary	3	MIKE SHE built-in Constants	1	MIKE SHE built-in Forcings	0	Derived Output

Figure 6.1 Enable MIKE ECO Lab for Unsaturated zone to incorporate the PGM into MIKE SHE.

After selection of the template, the GUI will show the summary of the template, i.e. the number of state variables, user defined constants, and if there are any MIKE SHE built-in constants (e.g. soil moisture) or MIKE SHE built-in forcings (e.g. solar radiation) which are used as inputs for the PGM (see Figure 6.2). The rest of the user defined constants and forcings have to be specified later under '*Unsaturated Flow*' tab (see Section 7) for unsaturated zone template, and similarly for the '*Saturated Zone*' template (see section 8).



Constant Name	MIKE SHE Parameter	MIKE SHE Parameter Data Type	Unit used in MIKE ECO Lab template
d_cell_top	☒	Depth to cell top	meter
d_cell_bottom	☒	Depth to cell bottom	meter
density	☒	Primary bulk density	kg/meter ³
POR	☒	Saturated water content	percent

Forcing Name	MIKE SHE Parameter	MIKE SHE Parameter Data Type	Unit used in MIKE ECO Lab template
Temp_she	☒	Air temperature	degree Celsius
Temp_soil	☒	Soil temperature	degree Celsius
ios_she	☒	SW Solar radiation	watt/meter ²
NH4_dep	☐		
NO3_dep	☐		
SM	☒	UZ water content	percent

Figure 6.2 Examples of MIKE SHE built-in constants and forcings for the 'Unsaturated Zone'.

Constant Name	MIKE SHE Parameter	MIKE SHE Parameter Data Type	Unit used in MIKE ECO Lab template
knit	☐		
tetr	☐		
hmt	☐		
mso3	☐		
kden	☐		
ksb	☐		
hno3	☐		
POR	☒	Primary porosity	()
d_cell_top	☒	Depth to cell top	meter
d_cell_bottom	☒	Depth to cell bottom	meter
k_redox	☐		

Forcing Name	MIKE SHE Parameter	MIKE SHE Parameter Data Type	Unit used in MIKE ECO Lab template
Temp_soil	☒	Soil temperature	degree Celsius

Figure 6.3 MIKE SHE built-in constants and forcings for the 'Saturated Zone'.

6.1 Species

The 'Species' tab reflects the number of state variables available in the MIKE SHE model. Once the PGM is imported into MIKE SHE, the 'Species' list is updated with the state variables in the PGM. Users can add more variables, for example, here, an existing state variable for Bentazone was already present before the import of PGM. In this case, Bentazone is not part of the PGM which means that Bentazone does not go through the processes included in the PGM. Bentazone would undergo other processes already defined in the original MIKE SHE model without PGM.

It should be noted that most of the 'Species' should be changed to 'Fixed (undefined)' type except for NH₄, NO₃, and DO for the PGM. These three state variables represent the dissolved NH₄, NO₃ and oxygen in the soil water matrix, which will be moved with the water movement calculation, and hence defined as 'Dissolved'.



Species					
	Include	Name	Type	ET Uptake Factor	Solubility in Surface Water
1	☒	Bentazone	Dissolved	1	1e+08
2	☒	BC1a_2D	Fixed (undefined)	0	1e+08
3	☒	BN1a_2D	Fixed (undefined)	0	1e+08
4	☒	BC1b_2D	Fixed (undefined)	0	1e+08
5	☒	BN1b_2D	Fixed (undefined)	0	1e+08
	Include	Name	Type	ET Uptake Factor	Solubility in Surface Water
41	☒	NH4	Dissolved	0	1e+08
42	☒	NO3	Dissolved	0	1e+08
43	☒	DO	Dissolved	0	1e+08

Figure 6.4 Define and check data types for all species.

6.2 Land Use

Different types of vegetation have their own specified timeseries defined under 'Land Use'. This is done for leaf area index (LAI), root depth (RD), crop coefficient (Kc) based on crop development stages. The PGM will calculate LAI, RD and Kc and transfer to MIKE SHE. Through this, vegetation development is calculated from the PGM instead of being prescribed.

Vegetation

☐ Define Vegetation Units (use *.etv-files)
 ☒ Define Parameters Individually

Figure 6.5 Under vegetation, define parameters individually for LAI, RD, Kc.

To enable the coupling of PGM and MIKE SHE, LAI, RD and Kc has to be specified individually (see Figure 6.5) using spatially distributed files, i.e. .dfs2 file format (see Figure 6.6). To produce these .dfs2 files for LAI, RD and Kc, refers to usage of the 'plant-growth-module' in the PDP toolbox. These are files with only 1 data point, hence note to untick 'Time varying'.

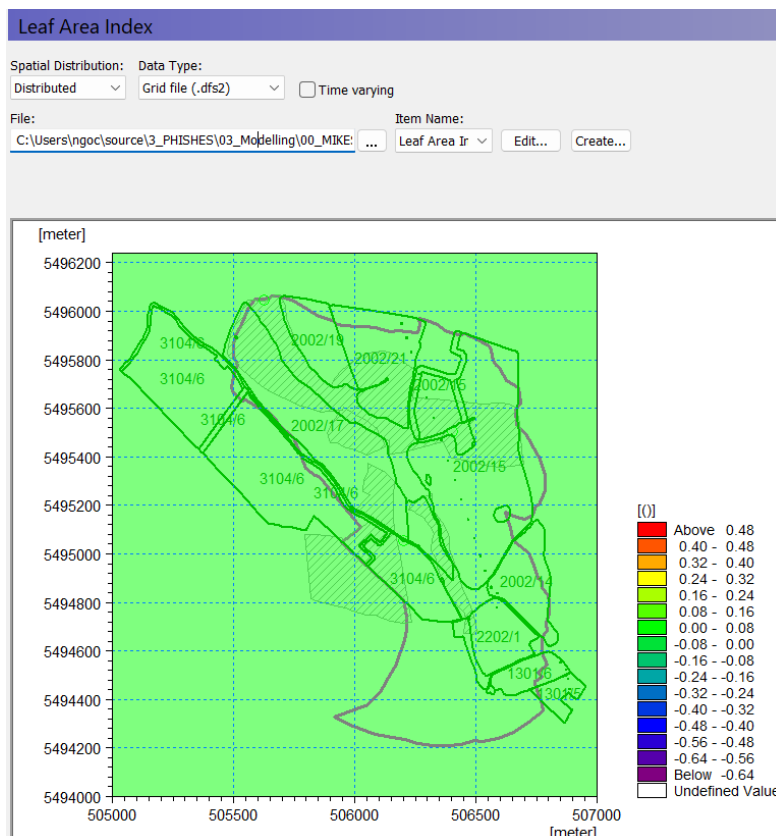


Figure 6.6 Example of specification for LAI. Note to untick 'Time varying'.



7 Unsaturated zone: Input preparation

As the catchment has varying land-use with each plot of land corresponding to a different plant functional group, the constants for the ECO Lab template must be defined as spatially varying.

After importing the ECO Lab template for unsaturated zone, users need to define 3 sections: constants, forcings (if any) and initial values (if any). The list of constants for the PGM model is provided Reference Technical Guide (Kuusemaa et al, 2026).

7.1 Usage of PDP plant-growth-module toolbox

7.1.1 Parameters based on soil and land use

As the PGM includes processes related to aboveground (e.g. crop/tree growth) as well as belowground (e.g. soil processes) which can be related to both land use and soil profile, the initial values and constants can be prescribed based on land use and soil profile accordingly.

Constants and initial values are generated using the `pgm_initial_condition_dfs2_map_generator.ipynb` from the 'MSHE-Ecolab-PGM'. This notebook has 4 components with a total of 6 files. Users have the option to specify initial values and constants based on land used and/or soil profile depending on the catchment model. The `pgm_initial_condition_dfs2_map_generator.ipynb` generates .dfs2 files for all constants and initial values specified.

Table 7.1 Files needed for `pgm_initial_condition_dfs2_map_generator.ipynb`.

Components	Files required	Files source
Land use	Land use definition: spatial data (.dfs2 file)	From 'Processed Data' ¹
	Land use classification mapping (.csv file)	User-defined (see Land use definition)
Soil profile	Soil profile definition: spatial data (.dfs2 file)	From 'Processed Data'
	Soil profile classification mapping (.csv file)	User-defined (see Soil profile definition)
Constants	All spatially varying constants (.csv file)	User-defined (see Constants)
Initial values	All spatially varying initial values (.csv file)	User-defined (see initial values)

- Land use definition

Figure 7.1 shows the grid code values for land-use in the .dfs2 files and Table 7.2 shows the land use code in .csv file for the Cernici catchment. The land-use code has to be unique.

Table 7.2 Content for .csv file for land use code. 'Apply' refers to whether the land use code is available and to be modelled within PGM.

CLASS	CODE	Apply
Mixture for nectar-producing fallow (eco-payment)	1	0
Clover-grass mixture (with a predominance of clovers)	2	1

¹ The 'Processed Data' tab in MIKE SHE GUI is the processed data after running the 'Preprocessing' step. This is where MIKE SHE scans through the model set up and interpolates all spatial data to the specified model grid (DHI, 2025). Refer to Appendix A.3 for example.



CLASS	CODE	Apply
Winter rapeseed	3	0
Winter rapeseed	4	0
Winter wheat	5	1
Corn for silage	7	0
Corn for silage	9	0
Winter rapeseed	10	0
Winter rapeseed	12	0
Spring barley	16	1
Trees and shrubs	17	0
Temporary grassland	19	0

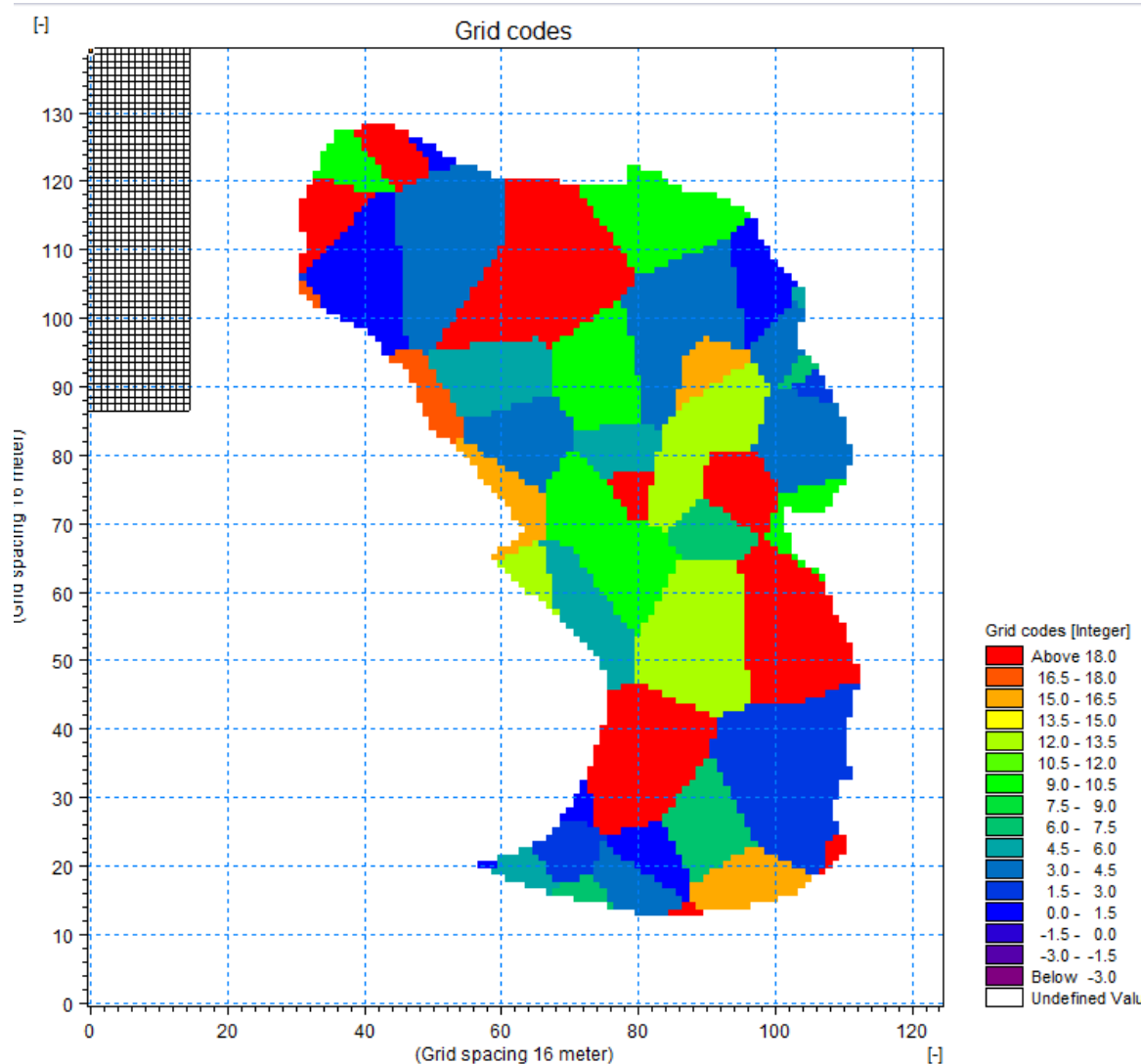


Figure 7.1 Land use definition presented in .dfs2 file.

- **Soil profile definition**

Figure 7.2 shows the grid code values for soil profile type in the .dfs2 file and Table 7.3 shows the soil profile code in .csv file for the Cernici catchment.



Table 7.3 Content for .csv file for soil profile code.

CLASS	CODE
Agricultural Soil	1
Forest Soil	2
Sandy	3
Loamy	4

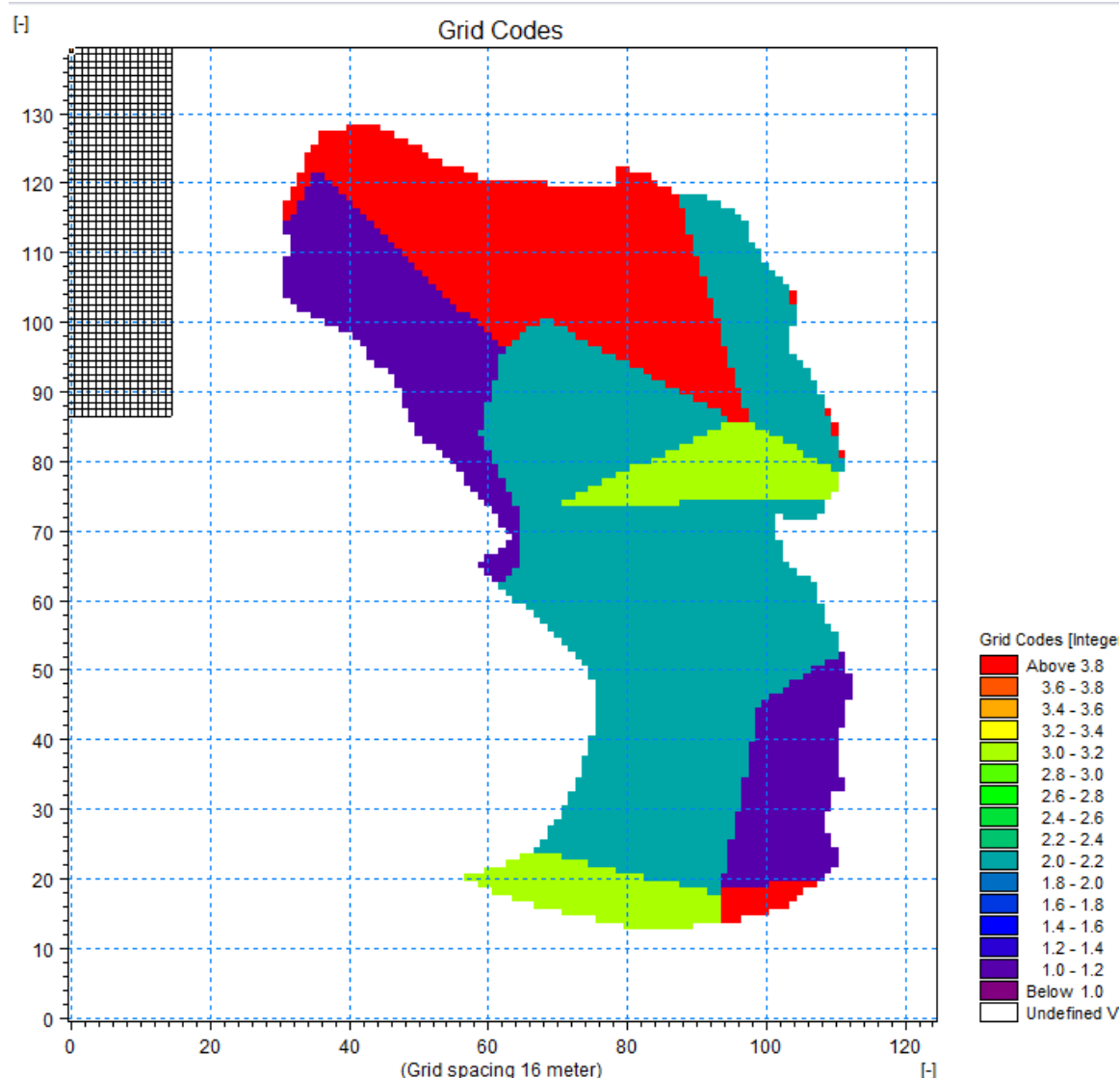


Figure 7.2 Soil profile definition presented in .dfs2 file.

• Constants and initial values

The constants for plant functional groups are based on the land use. For example, the maximum growth rate (mybc1) is specified only for functional group 1 (spring barley). On the other hand, there are other constants such as mineralisation rate organic matter in soil (krsoc) can be dependent on soil type. As such, the master-list (.csv file) specifying the constants used in PGM would require:

- o which profile are the constants based on, i.e. land use or soil profile



- o which profile ID the constants should be applied to, e.g. the name of the land use or the soil code
- o the constants' names
- o the constants' values
- o whether the constants need to be applied as map, i.e. .dfs2 file

The *pgm_initial_condition_dfs2_map_generator.ipynb* takes in the specified .csv file and output a number of .dfs2 files. An example of the .csv file is presented in the Table 7.4.

The same template .csv file can be used for initial values.

Table 7.4 Example content for .csv file for constants

Template	SpeciesID	Parameter	Value	Type
soilprofile	1	krsoc	0.002	1
soilprofile	2	krsoc	0.003	1
soilprofile	3	krsoc	0.004	1
soilprofile	4	krsoc	0.005	1
landuse	Clover-grass mixture (with a predominance of clovers)	tmax	22	1
landuse	Winter wheat	tmax	22	1
landuse	Spring barley	tmax	16	1
landuse	Mixture for nectar-producing fallow (eco-payment)	tmax	25	1
landuse	Clover-grass mixture (with a predominance of clovers)	topt	18	1
landuse	Winter wheat	topt	18	1
landuse	Spring barley	topt	14	1
landuse	Mixture for nectar-producing fallow (eco-payment)	topt	22	1

7.1.2 Field capacity and wilting point

Field capacity (Fc) and wilting point (Wp) are soil properties. These soil properties vary spatially and vertically in the soil column. These two constants can be prepared using the *pgm_soil_profile_setup.ipynb* from the 'MSHE-Ecolab-PGM' which generates .dfs2 files for each soil layer in the unsaturated zone.

The inputs for the *pgm_soil_profile_setup.ipynb* requires outputs from the 'Preprocessing' step in MIKE SHE. The files are in the user-specified outputs folder when running 'Preprocessing' step in MIKE SHE:

- _PreProcessed.dfs2 file
- PreProcessed_SoilProf.txt files containing soil properties information for each soil grid code

The .dfs2 file for Soil Profile Grid code can also be generated from 'Processed Data' tab from the MIKE SHE GUI (see Appendix A.3).

7.1.3 Forcings

Besides the forcings supplied from MIKE SHE, other forcings e.g. seeding for crops, fertilization can be specified wherever necessary. The forcings are generated using the *pgm_forcing_generator.ipynb* from the 'MSHE-Ecolab-PGM'. These forcings are specifically based on landuse only. The inputs required are:

- .dfs2 file for landuse grid code



- .yaml file (see Figure 7.3)

The input for the `pgm_forcing_generator.ipynb` is contained in a .yaml file with the following format:

```
timeseries_inputs:
  var1:
    output_grid: Jytland-test-ouput.dfs2
    item_name: Seed Application Rate
    inputs:
      - grid_code: 1
        path: sample_data/pgm_forcing_generator/example_dfs0_dfs2/DMI_Jutland_2011-2023_item1.csv
      - grid_code: 5
        path: sample_data/pgm_forcing_generator/example_dfs0_dfs2/DMI_Jutland_2011-2023.dfs0
        item: 5

  var2:
    output_grid: Jytland-test-ouput-var2.dfs2
    item_name: Fertilizer Application Rate
    inputs:
      - grid_code: 1
        path: sample_data/pgm_forcing_generator/example_dfs0_dfs2/DMI_Jutland_2011-2023_item2.csv
      - grid_code: 5
        path: sample_data/pgm_forcing_generator/example_dfs0_dfs2/DMI_Jutland_2011-2023.dfs0
        item: 1
```

Figure 7.3 Example of .yaml file specifying the forcings time series, either in .csv file or .dfs0 file. The grid code corresponds to the landuse grid code. The .csv file contains only 1 time series. The .dfs0 file can contain multiple time series, and users must specify the column number, i.e. 'item'. For example, 'item: 5' means 5th column (or 5th item) in the .dfs0 file.

Users then must specify the details of the forcings properties in 'Step 2: Configure Forcings' in the '`pgm_forcing_generator.ipynb`', see example in Figure 7.4. Mainly, the path to the .yaml file needs to be specified.

The output is a .dfs2 file which has time varying to be used for forcings in MIKE SHE.

```
# All forcings are defined in a single YAML config: one named entry per forcing,
# each with its own output DFS2, item metadata and per-grid-code timeseries.
# Shared settings (grid-code DFS2, output folder, EUM defaults) live under
# 'defaults:' in the same file. Grid codes present in the grid-code DFS2 but not
# listed for a forcing are automatically filled with a constant value of 0.
FORCING_CONFIG = MODULE_ROOT.joinpath(
    "sample_data",
    "pgm_forcing_generator",
    "timeseries_inputs.yaml",
)

FORCINGS = forcing_generator_native.load_forcing_configs(MODULE_ROOT, FORCING_CONFIG)

# Stop at the first failing forcing, or run the rest and report failures at the end.
CONTINUE_ON_ERROR = False

print("Forcing config (YAML): ", FORCING_CONFIG)
print("Number of forcings: ", len(FORCINGS))
for forcing in FORCINGS:
    print(
        f" - {forcing['name']}: {len(forcing['timeseries_inputs'])} timeseries input(s)"
        f" -> {forcing['output_grid']}"
    )
```

Figure 7.4 Example of specifying the details of the forcings properties.



- **Note: Simulations with Crop Rotation**

The user is required to provide a time varying map of 0 or 1 that indicates grass removal. This map is inserted under “Unsaturated Flow – MIKE ECO Lab Forcings” in the MIKE SHE set up panel. In that same setup section, users need to indicate, for example, seeding trigger at the same time or near grass removal timing. This is because seeding can potentially happen earlier than the removal of grass trigger. The current model is set up such that there are conditions that other biomasses need to be removed before the growth of a different crop can start.

7.2 Set up in MIKE SHE

7.2.1 Initial values and constants

Example of spatially varying constant of maximum growth rate for spring barley (mybc1) specified in MIKE SHE is shown below. When specifying using spatially varying option, users must choose ‘Layer based’ under ‘Spatial Distribution’ (see Figure 7.5) and add 1 layer. Subsequently, users must specify the lower level at which the constant is applied to, i.e. the entire depth of the unsaturated zone’s soil profile (see Figure 7.6) and the corresponding .dfs2 file (see Figure 7.7).

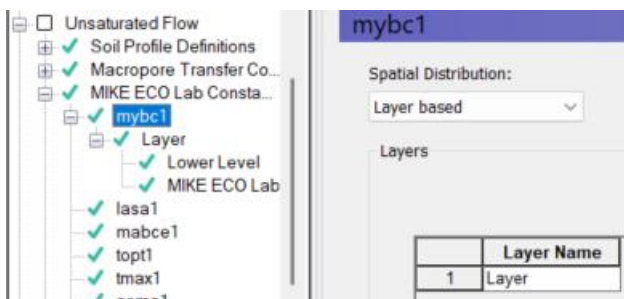


Figure 7.5 Specify spatial distribution of the constant: Layer based

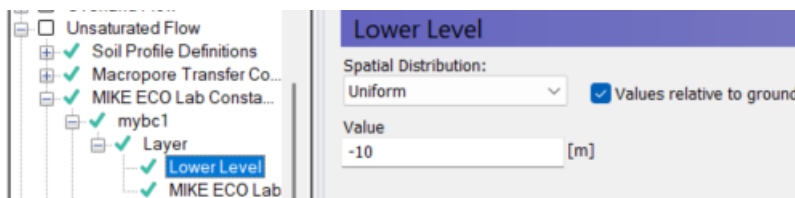
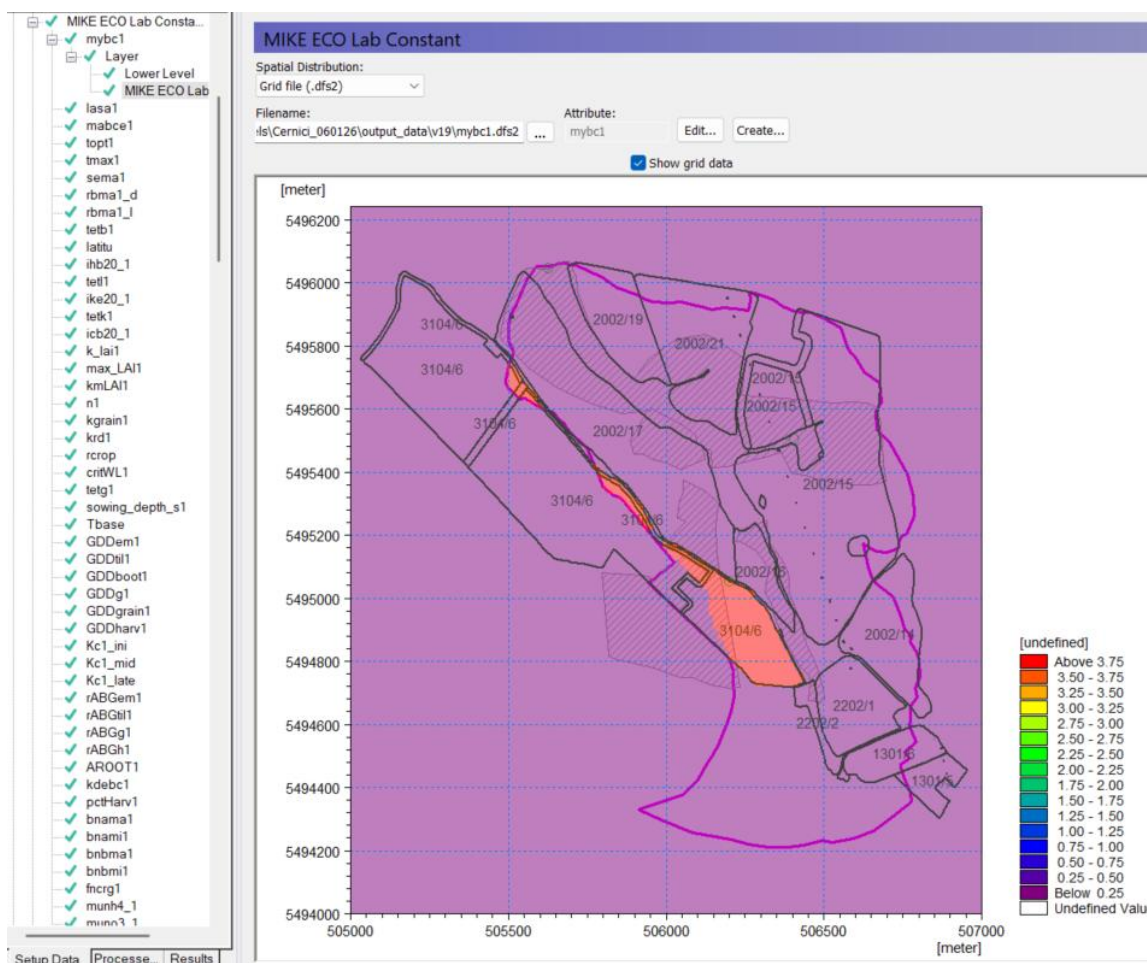


Figure 7.6 Specify the lower level until which the constant is applied to. The lower level should correspond to the thickness of the soil layers for unsaturated zone.



7.2.2 Field capacity and wilting point

These values for field capacity and wilting point are unchanged unless the soil profile definition for the unsaturated zone is changed. As such, users only need to utilize the *'pgm soil profile setup.ipynb'* once and set up Fc and Wp in MIKE SHE's GUI once.

7.2.3 Forcings

The forcing are the time series, spatially varying *.dfs2* files. As such, similar to the setup from section 7.2.1, the layer based spatial distribution must be selected with defined lower level and corresponding *.dfs2* files.



8 Saturated zone: Input preparation

At this stage, for the saturated zone, there are no processes related to vegetation/crops. The only processes available are related to denitrification. As such, there is no need to include spatially varying constants and forcings for the saturated zone; the constants are setup as '*Uniform*'.



9 Model Execution

Before running the model, users need to make sure that there are no errors in the setup. Errors are 'red cross-marks' which appear at any setup tabs if there is an error.

After making sure there is no error in the setup, the user must go to the folder where the MIKE SHE setup *.she* file e.g. *Cernici.she*, make a copy of the *.she* file and rename as *Cernici_AdCou*. Note that the *_AdCou* file does not have any file extension. This file allows for the coupling between MIKE SHE and MIKE ECO Lab. See section 11.1 for errors occurred if *_AdCou* file is missing.

Subsequently, users need to run the *Preprocessing* (PP) step followed by simulation with *water movement* (WM).



10 Result Inspection

The current result inspection script 'obs_res_compare_cernici.ipynb' takes the .dfs3 results from the model simulation to compare with observed data in time series plots. Currently, the script is based on the available observed data from the Cernici model which currently includes: actual evapotranspiration, harvested biomass, LAI, root depth and porewater concentration of nitrogen fractions (i.e. NH₄, NO₃) (see Figure 10.1). Users can use this script as a guide to modify for similar purposes of model results inspection.

Note that the measured data must be processed into separate folder corresponding to the parameters to be compared, and that each parameter is a standalone MIKE Zero data file: it can be time series data (.dfs0) or spatially and temporal varying data (.dfs2) files. See example in Figure 10.1 where there are 2 MIKE Zero Time Series (.dfs0) for ETa (actual evapotranspiration) and ETr (reference evapotranspiration) in the ET folder and a MIKE Zero .dfs2 file for spatially and temporally varying LAI. Users can add more measured data in the same manners whenever necessary.

ET	4/24/2026 8:46 AM	File folder
GWL	4/24/2026 9:16 AM	File folder
Harvest_dry_grain	4/7/2026 11:57 AM	File folder
LAI	4/7/2026 10:55 AM	File folder
Root Depth	4/27/2026 4:36 PM	File folder
Soil_water_N_P	4/28/2026 3:42 PM	File folder
ETa.dfs0	4/7/2026 4:35 PM	MIKE Zero Time Series
ETr.dfs0	1/27/2026 2:39 PM	MIKE Zero Time Series
LAI_Mart_200_1981-2020.dfs2	4/8/2026 4:00 PM	MIKE Zero Data Manager

Figure 10.1 Example of processed measured data before using the result inspection script.

First, users have to first specify the file paths of measured data as well as the simulation results (see Figure 10.2).

```
# User-defined arguments

# Directories
Obs_dir = pathlib.Path(r"\\DKCPH1-STOR.DHI.DK\Projects\11829965\WP1_PGM\ObservedData\Processed")
Res_filepath = pathlib.Path(r"C:\Users\almm\OneDrive - DHI\Desktop\Cernici16_Ben_v101_WM_AD_ECOLabv15_PDPTools_WQ_3DUZ.dfs3")

#Output directory default same as results
Output_dir = Res_filepath.parent / "ObsResCompare"
```

Figure 10.2 File path of measured data and simulated results.

Subsequently users can define parameters to compare. Figure 10.3 shows an example of possible parameters to compare based on available measured data from Cernici. For example, 'ET2compare' is a variable in the script where two measured data is available and modelled: 'Actual Evapotranspiration' is the user-defined name of the parameter of interest; 'ETa' is the name of the modelled results which is pre-defined in the PGM; 'ETa' is the name of the measured data.

It should be noted that the name of the pre-defined modelled results should not be changed and is case-sensitive. The list of modelled state variables are available in the ECO Lab template for the PGM as well as at the 'Species' list in MIKE SHE's GUI (see section 6.1). The name of the



measured data in the result inspection script should be the same as the name specified in the properties of the MIKE Zero file (see Figure 10.4).

```
obs2compare = [
    'ET',
    # 'GWL',
    'Harvest_dry_grain\\BCc',
    'LAI',
    'Root Depth',
    'Soil_water_N_P'
]

# Dicts for comparing model results to the observations of each type
# dict entry = "your name key": ["Results state variable name", "Observations file name without extension"]
# 'Harvest_dry_grain\\BCc'
crops2compare = {'Spring Barley': ["BC1c", "Spring barley_BCC"],
                 'Grass': ["BC2c", "Permanent grassland_BCC"],
                 'Winter Barley': ["BC3c", "Winter barley_BCC"],
                 }

# 'ET'
ET2compare = {'Actual Evapotranspiration': ["EtA", "ETa"],
              'Reference Evapotranspiration': ["RefEt", "ETr"]
              }

# 'LAI' and 'Root Depth'
eo2compare = {'LAI': ["LAI", "LAI_Mart_200_1981-2020"],
              'Root Depth': ["RD", "Rd_Mart_200_1981-2020"]
              }

# 'Soil_water_N_P'
porewater2compare = {'Porewater Column NH4': ["S_mNH4", "Suction Cups"],
                     'Porewater Column NO3': ["S_mNO3", "Suction Cups"],
                     'Groundwater Column NH4': ["S_mNH4", "Lysimeter"],
                     'Groundwater Column NO3': ["S_mNO3", "Lysimeter"],
                     'Porewater Layer NH4': ["NH4", "Suction Cups"],
                     'Porewater Layer NO3': ["NO3", "Suction Cups"],
                     'Groundwater Layer NH4': ["NH4", "Lysimeter"],
                     'Groundwater Layer NO3': ["NO3", "Lysimeter"]
                     }

# Extra user defined arguments for porewater
lysimeter_layer = 0 # bottom
suctioncup_layer = 23 # top
```

Figure 10.3 Modelled state variables to compare with measurements in Cernici model.



File Properties

General Information

Title:

Axis Information

Axis Type:

Start Time:

Time Step: [days]
 [hour:min:sec]
 [fraction of sec.]

No. of Timesteps: Axis Units:

Delete Value

Single Precision:
Double Precision:

Convert Precision for all Items

Convert to:

Item Information

	Name	Type	Unit	Precision	TS Type
1	ETa	Evapotranspiration Rate	mm/day	Single	Instantaneous

Figure 10.4 Example: File properties for measured ETa. Choose the appropriate Type and Unit. If not available, users can choose Undefined Type and Undefined Unit.



11 Common errors

11.1 Missing *_AdCoup* file

Before executing the model, users need to make a copy of the *.she* setup file and rename it to *_AdCoup* (without any extension). This allows for coupling between MIKE SHE and the PGM. If the *_AdCoup* file is missing, the following error, see Figure 11.1, would appear in the GUI.

```
raise Exception("Error initializing WQ")
Exception: Error initializing WQ

Error while emitting signal "postEnterSimulator"

Termination...
```

Figure 11.1 Error when missing *_AdCoup* file.



12 References

DHI. (2025). MIKE ECO Lab - Numerical Lab for Ecological and Agent Based Modelling - User Guide.

DHI. (2025). MIKE SHE - User Guide and Reference Manual.

Kuusemae, K., Lørup, J. K., Leirião, S., Murray, A., Thai, N. B. (2026). *Plant Growth Module - Technical Reference Guide*. DHI. – awaiting publication.



APPENDICES

Appendix A Supporting Tools

Appendix A.1 De-bugger quick guide

The Python script '*plot_mikeshe_veg_cernizi_sz.py*' is a script to investigate the simulation results should users wish to look into the results of individual processes calculated in the PGM at a specific location in the model.

The script includes specifying the file path of the simulation results '*iData*'. The outputs are a series of images (.png files) which include contour plot (i.e. temporally varying across the entire soil column), time series of the first 4 soil layers, etc. for parameters and auxiliary calculations in the PGM.

The script requires specifying the location of result extraction by specifying the cell number for '*loc_x*' and '*loc_y*'.

```
iData=r'Cernici16_Ben_v101_WM_AD_ECOLabv15.she - Result Files'
iRes=r'Cernici16_Ben_v101_WM_AD_ECOLabv15.she - Result Files\png'

files=[f for f in os.listdir(r'{}'.format(iData)) if '.dfs3' in f]
# files3D=['WmWqBox_5x6_KADR_v10_UZ_SZ_WQ_3DUZ.dfs3', 'WmWqBox_5x6_
# files2D=['WmWqBox_5x6_KADR_v10_UZ_SZ_ET_UzCells.dfs2']

no_UZ_layer = 24
max_layer=no_UZ_layer-1
loc_x, loc_y = 100, 100
```

Figure 12.1 Debug script: Users input simulation results folder and location of result extraction.



Appendix A.2 WQ hotstart

Currently, MIKE SHE only allows for saving hotstart for states related to the water movement processes. A tool was developed to allow for extraction of PGM's results at the end of one simulation to be used as initialized values for another simulation, i.e. a hotstart for states related to PGM. This will allow for debugging purposes and start-stop type of simulations. The tool used is 'pgm_initial_condition_updater.ipynb'.

The tool requires 2 input files: 1) the .she model file and 2) the .dfs3 for the PGM's results. The model output .she model setup file with initialized values at each UZ soil layer for all state variables in PGM.

```
# User-editable parameters
MAIN_DIR = Path(
    r"C:\Users\ngoc\source\3_PHISHES\phishes-pdp\plant-growth-module\sample_data\initial-value-updater"
)

DFS3_PATH = MAIN_DIR.joinpath(
    "Cernici16_Ben_v101_WM_AD_v20_v1_ALMM_KADRparams.she - Result Files",
    "Cernici16_Ben_v101_WM_AD_v20_v1_ALMM_KADRparams_WQ_3DUZ.dfs3",
)
SHE_INFILE = MAIN_DIR.joinpath(
    "Cernici16_Ben_v101_WM_AD_v20_v1_ALMM_KADRparams.she"
)
OUTPUT_DIR = MAIN_DIR.joinpath("processed")

TIMESTEP = -1
REVERSE_Z = True
SPECIES = None # None = all matched species

print("DFS3:      ", DFS3_PATH)
print("SHE infile:", SHE_INFILE)
print("Output dir:", OUTPUT_DIR)
```

Figure 12.2 Inputs for the toolbox to set up .she with hotstart from PGM's results the MAIN_DIR (main directory) contains the .she model setup file and the folder containing WQ_3DUZ.dfs3 PGM's results file. The model output .she model setup file with initialized values at each UZ soil layer.



Appendix A.3 Extract processed landuse and soil grid code

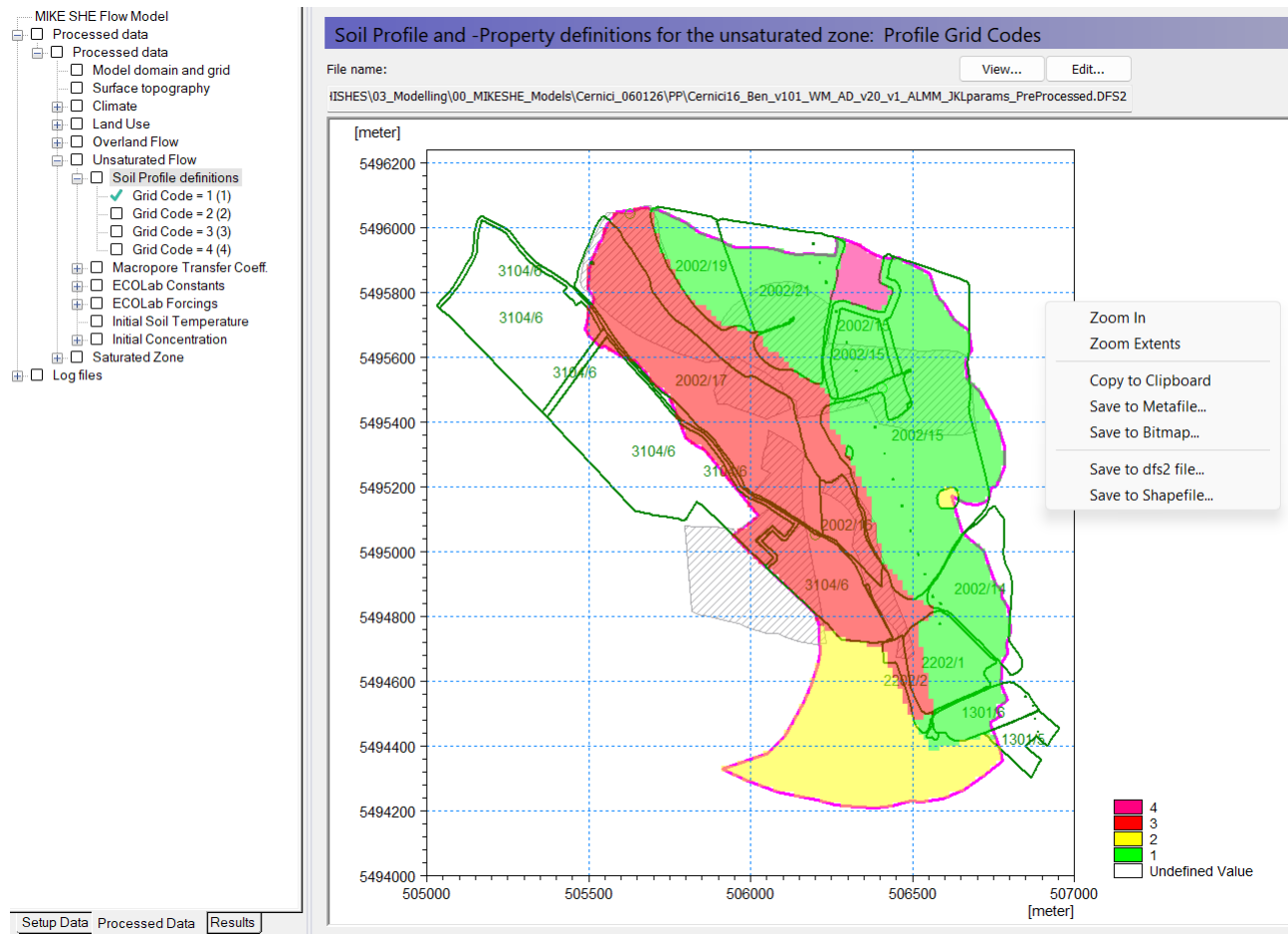


Figure 12.3 Extraction of soil grid code from processed data in MIKE SHE GUI.

The .dfs2 file for Soil Profile Grid code can also be generated from 'Processed Data' > 'Soil Profile definitions' tab from the MIKE SHE GUI. Similarly for landuse grid code, users can extract the .dfs2 file from 'Processed Data' > 'Land Use' from the MIKE SHE without PGM processed data.

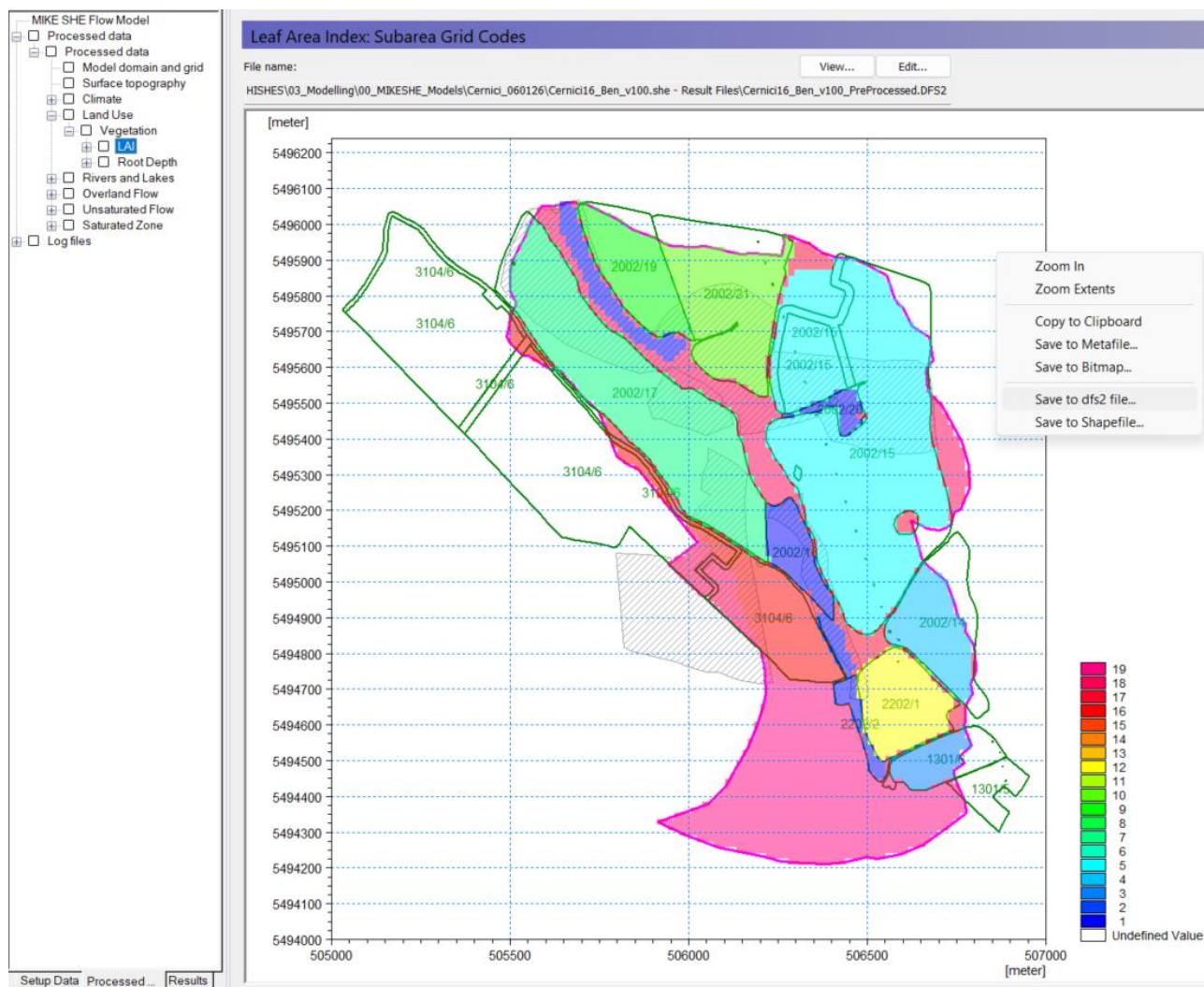


Figure 12.4 Extraction of landuse grid code from processed data in MIKE SHE GUI.

For More Information

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